

Refining the admissible competitor class $\mathcal{A}_{\text{adm}}$: four exclusion boundaries,

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

The H-LAYER consolidation reduced the remaining H-LAYER item to the completeness of the admissible competitor class $\mathcal{A}_{\text{adm}}$. v1.0 examined three exclusion boundaries and concluded the sign-off reduces to “crystalline order”. The operator adversarial review found a LATTICE-but-OFF-SHELL escape that v1.0 did not handle. This re-issue corrects the $\mathcal{A}_{\text{adm}}$ definition, adds the missing boundary, and records the universal fallback under which the analytic thresholds survive. Scope: operating endpoint $I = 2 \times 10^{-3}$.

2. The corrected $\mathcal{A}_{\text{adm}}$ definition

$$\mathcal{A}_{\text{adm}} = \left\{ Q \subset \bigcup_{R \in \mathcal{R}} (\Lambda \cap \{|x|^2 = R\}) : \begin{array}{l} |Q| \leq n_{\text{pack}}, \quad Q = -Q, \\ \angle(k_i, k_j) \geq \theta_{\min}(\hat{r}), \\ Q \text{ a registered crystallographic shell or shell-union competitor} \end{array} \right\}.$$

The pin $T'(Q) \leq 13$ is asserted ONLY on this registered shell / shell-union competitor class. For arbitrary antipodal lattice subsets the fallback is only $T'(Q) \leq N/2 \leq 20$.

3. The four exclusion boundaries

Boundary 1 – sub- $\theta_{\min}$ (DERIVED). The coherence-indistinguishability lemma bounds sub-resolution restructuring, $|F[P'] - F[P]| \leq c_{\text{ind}} I^2 = \text{MARGIN}/33$ at the endpoint ($c_{\text{ind}} \approx 30\text{--}32$, $\theta_{\min} = 0.627$); sub- $\theta_{\min}$ modes are absorbed into the Gaussian sea.

Boundary 2 – super- n_{pack} (DERIVED). The spherical packing bound gives $\leq 16/\theta_{\min}^2 = n_{\text{pack}} = 40.7$ separated modes; $> n_{\text{pack}}$ forces a pair below $\theta_{\min}$ (pigeonhole), reducing to Boundary 1.

Boundary 3 – lattice but off-shell / arbitrary multi-radius subsets (FALLBACK-CONTROLLED). The direct adversarial construction shows the sharp pin $T'(Q) \leq 13$ is NOT a theorem over arbitrary antipodal lattice subsets: there exist antipodal lattice sets with

$$N = 40, \quad T'(Q) = 20 > 13.$$

These are not non-lattice; they are lattice-supported but not registered single-shell or shell-union crystallographic competitors. However, any real antipodal competitor obeys the universal geometric fallback

$$T'(Q) \leq \frac{N}{2}$$

(the sum-circles C_t and C_{-t} are disjoint for $t \neq 0$, and antipodal symmetry maps $Q \cap C_t$ onto $Q \cap C_{-t}$, so $2T' \leq N$). With real antipodal symmetry and $N \leq n_{\text{pack}} = 40.7$, hence $N \leq 40$,

$$T'(Q) \leq 20, \quad 20 < 20.5 < 26.2 < 60.4.$$

Thus arbitrary antipodal escape competitors can **FALSIFY** the sharp $T' \leq 13$ pin, but they do NOT falsify the H-diag (RES-1), RES-5, or SC-SCOPE thresholds, provided the packing bound $N \leq 40$ and real antipodal symmetry are retained. The cost is thinness: at $T' = 20$ the binding RES-1 ratio is $R_{\text{lead}}(20) = 0.974 (\times 1.026)$, still < 1 ; the SC-SCOPE joint is $\times 1.082$, RES-5 floor $\times 1.31$. *Whether such off-shell escape competitors are physically admissible in B1/B2 is the remaining operator question.*

Boundary 4 – non-lattice / incommensurate (MODELLING, with fallback). A non-crystalline condensate is a different ansatz: the crystalline-order restriction is the genuine modelling input, softened by DR-2 decoupling $T' \ll_{\varepsilon} N^{\varepsilon}$ (R-022 T6-conditional / R-023 T4), off the critical path.

4. Verdict

Of the FOUR boundaries: 1 and 2 are physically DERIVED; 3 is FALLBACK-CONTROLLED ($T' \leq N/2 \leq 20$, thresholds survive thinly); 4 is a MODELLING input. The operator completeness sign-off therefore reduces to “the ground state is crystalline AND shell-supported” (a registered shell / shell-union competitor), NOT merely crystalline. No tier flip: B1/B2 T6 on {H-LAYER}, B5 T5.

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “v1.0 over-claimed by conflating the escape with non-lattice.” UPHELD and corrected: the escape is LATTICE but off-shell (Boundary 3), a distinct case v1.0 folded into “non-lattice”. v1.1 separates them and adds the $T' \leq N/2$ fallback.
- (β) “ $R_{\text{lead}}(20) = 0.974$ is razor-thin – RES-1 nearly fails.” ACKNOWLEDGED: the off-shell fallback thins RES-1 to $\times 1.026$. It does not break it, but the comfortable margins ($\times 1.54$ etc.) hold only on the registered $T' \leq 13$ class. This is exactly why the shell-supported restriction matters.
- (γ) “Does anything flip?” DISMISSED: no. The thresholds survive the fallback; the sign-off is sharpened (crystalline + shell-supported); B1/B2 stay T6.

Quantitative sanity check (mandatory): *Boundary 1* $c_{\text{ind}} I^2 = \text{MARGIN}/33$; *Boundary 2* $n_{\text{pack}} = 40.7$; *Boundary 3* $T' \leq N/2 \leq 20$ (verified on antipodal samples), $R_{\text{lead}}(20) = 0.974 < 1$, $\text{joint}(20) = \times 1.082$, $20 < 20.5 < 26.2 < 60.4$; *Boundary 4* $T' \ll N^{\varepsilon}$.

6. Result footer

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Result ID:          B5-BEYOND-LAYER-BOUND / A_adm
                    exclusion-boundary refinement (v1.1)
Precise statement:  A_adm = {Q subset shell-UNION : |Q|<=n_pack,
                    Q=-Q, angle>=theta_min,
                    registered shell/shell-union}. FOUR boundaries:
                                                              (1)
                                                              sub-theta_min

                    DERIVED (coherence lemma, MARGIN/33); (2)
                    super-n_pack DERIVED
                    (packing 40.7, pigeonhole); (3)
                    lattice-OFF-SHELL FALLBACK-
                    CONTROLLED -- sharp T'<=13 fails (escape
                    N=40,T'=20) but universal
                    antipodal T'<=N/2<=20 survives all thresholds
                    (20<20.5<26.2<60.4;
                    RES-1 razor-thin R_lead(20)=0.974); (4)
                    non-lattice MODELLING +
                    DR-2 decoupling. Sign-off = crystalline +
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SHELL-SUPPORTED.

Scope: Operating endpoint $I=2e-3$. Sharp $T' \leq 13$ = registered shell/shell-union class only; fallback $T' \leq N/2 \leq 20$ for arbitrary antipodal.

Dependencies: coherence-indistinguishability-lemma v1.0; hlayer-analytic-closure-consolidation v1.0 (A_{adm}); res5-chi-link-bypass ($T' \leq 13$ pin); dr2-decoupling-closure (R-022) / dr2-affine-invariance (R-023).

Evidence grade: ANALYTIC (4-boundary grading + $T' \leq N/2$ proof) + EXECUTED (res5_023_aadm_exclusion_boundaries.py v1.1 5/5).

Reproduction command: python codes/vacuum/res5_023_aadm_exclusion_boundaries.py

Expected output: theta_min=0.627; n_pack=40.7; $T' \leq N/2$ on samples; $R_{\text{lead}}(20)=0.974$ <1; joint(20)=x1.082; 5/5 PASS.

Falsification gate: An antipodal lattice subset with $T' > N/2$ (breaks the universal fallback); or $N > 40$ with all pairs $\geq \text{theta_min}$ (breaks packing); or an off-shell escape with $N \leq 40$ and $T' > 20$ (would need $T' > N/2$).

Tier before / after: No tier flip. B1/B2 T6 on {H-LAYER}, B5 T5. The A_{adm} sign-off is sharpened to crystalline + shell-supported; off-shell escapes are fallback-controlled ($T' \leq 20$, RES-1 thin x1.026), not breaking.

No-overclaim statement: Does NOT prove A_{adm} complete and does NOT flip B1/B2. The sharp $T' \leq 13$ pin is class-specific (registered shell/shell-union); the universal fallback is $T' \leq N/2 \leq 20$, under which the thresholds survive thinly. The sign-off (crystalline + shell-supported) is the operator's.

Next required action: OPERATOR: decide whether off-shell / arbitrary multi-radius antipodal lattice subsets are physically admissible B1/B2 competitors. If excluded (shell-supported), the comfortable $T' \leq 13$ margins hold; if admitted, the RES-1 x1.026 fallback binds.